COMMERCE ORDER FULFILMENT DATA PLATFORM

End-to-End Data Engineering Pipeline using Python, SQLite, Apache Airflow and Power BI

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AGENDA

- Business Problem
- Objectives
- Dataset
- Technology Stack
- ETL Architecture
  - Bronz layer
  - Silver layer
  - Gold layer
- Airflow
  - SQLite
- Dashboard
  - Insights
- Conclusion

BUSINESS PROBLEM

📁Multiple CSV Files

📊 Manual Reporting

📦 Order Tracking Challenges

🚚 Delivery Monitoring

😊 Customer Satisfaction Analysis

Goal: Build an automated ETL pipeline to transform raw e-commerce data into analytics-ready insights.

OBJECTIVES

- 📥 Data Ingestion
- 🧹 Data Cleaning
- 🏗️ Gold Layer Creation
- ⚙️ Airflow Automation
- 🗄️ SQLite Storage
- 📈 Power BI Dashboard

Dataset Overview

Source: Olist Brazilian E-Commerce Dataset

Orders 99,441	Customers 99,441	Products 32,951
Payments 103,886	Reviews 99,224	Sellers 3,095

Tables Used:
- Orders
- Customers
- Products
- Payments
- Reviews
- Sellers
- Order Items

TECHNOLOGY STACK

Tools and Technologies Used Throughout the ETL Pipeline

🐍 Python - ETL Development

↓

🐼 Pandas - Data Cleaning & Transformation

↓

🗄 SQLite - Data Storage

↓

🌬 Apache Airflow - Workflow Orchestration

↓

📊 Power BI - Dashboard & Analytics

↓

🐙 GitHub - Version Control

Project Architecture

📂 Raw CSV Data

↓

🟫 Bronze Layer

↓

🧹 Silver Layer

↓

🥇 Gold Layer

↓

🗄️ SQLite DB

↓

🌬️ Apache Airflow

↓

📊 Power BI Dashboard

BRONZE LAYER

Raw Data Ingestion & Storage Layer

📂 Raw CSV Files (Orders, Customers, Products, Payments, Reviews, Items, Seller) → 🗄 Bronze Layer

🎯 **Purpose**

- 📦 Store Raw Data Ingestion
- 🔒 Preserve Original Data
- 📝 No Data Modification

⚙ **Key Activities**

- 📥 Raw CSV
- 📂 Historical Storage
- ✅ Source Integrity

SILVER LAYER

(Data Cleaning, Transformation & Validation)

Bronze Layer → 🧹 Cleaning Process → 🩶 Silver Layer

🎯 **Purpose**	⚙️ **Key Activities**
🧹 Clean Data	❌ Remove Duplicates
✔️ Improve Quality	🔄 Convert Timestamps
📊 Standardize Data	🚫 Handle Missing Values
🛡️ Validate Records	✅ Data Validation

💡 Clean, validated and standardized datasets are prepared for creating Gold Layer analytical tables.

GOLD LAYER

(Analytics-Ready Data Warehouse)

Silver Layer

Gold Layer

Fact Tables

Orders

Payments

Reviews

Delivery SLA

Dimension Tables

Customer

Product

Seller

Purpose

- Business Analytics
- KPI Reporting
- Performance Analysis
- Dashboard Creation

Output

- Star Schema
- SQLite Database Tables
- Power BI Ready
- Optimized for Fast Queries

DATA QUALITY

(Ensuring Data Accuracy, Consistency, and Reliability)

📋 Validation Checks

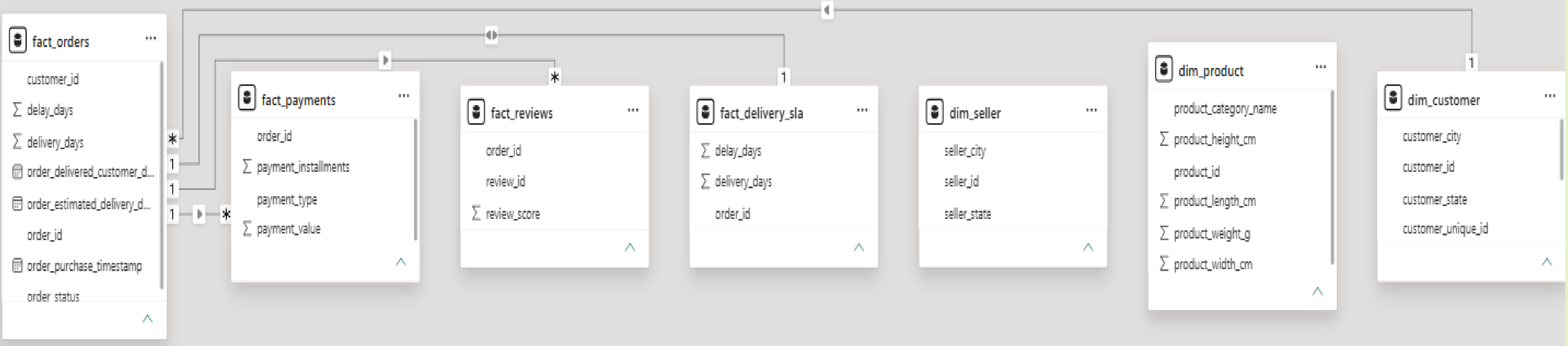
✅ Validation Rule	📈 Result
Duplicate Order IDs	Passed
Missing Values Handling	Completed
Timestamp Format Validation	Passed
Negative Payment Values	Passed
Negative Freight Values	Passed
Data Type Standardization	Completed

💡 Key Outcome

All validation checks were successfully completed, producing clean, consistent, and analytics-ready data for the Gold Layer and Power BI Dashboard.

STAR SCHEMA


Customer
Seller
Fact Orders
Product
Payments
Reviews
Delivery SLA


fact_orders
customer_id
delay_days
delivery_days
order_delivered_customer_d...
order_estimated_delivery_d...
order_id
order_purchase_timestamp
order_status
fact_payments
order_id
payment_installments
payment_type
payment_value
fact_reviews
order_id
review_id
review_score
fact_delivery_sla
delay_days
delivery_days
order_id
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seller_state
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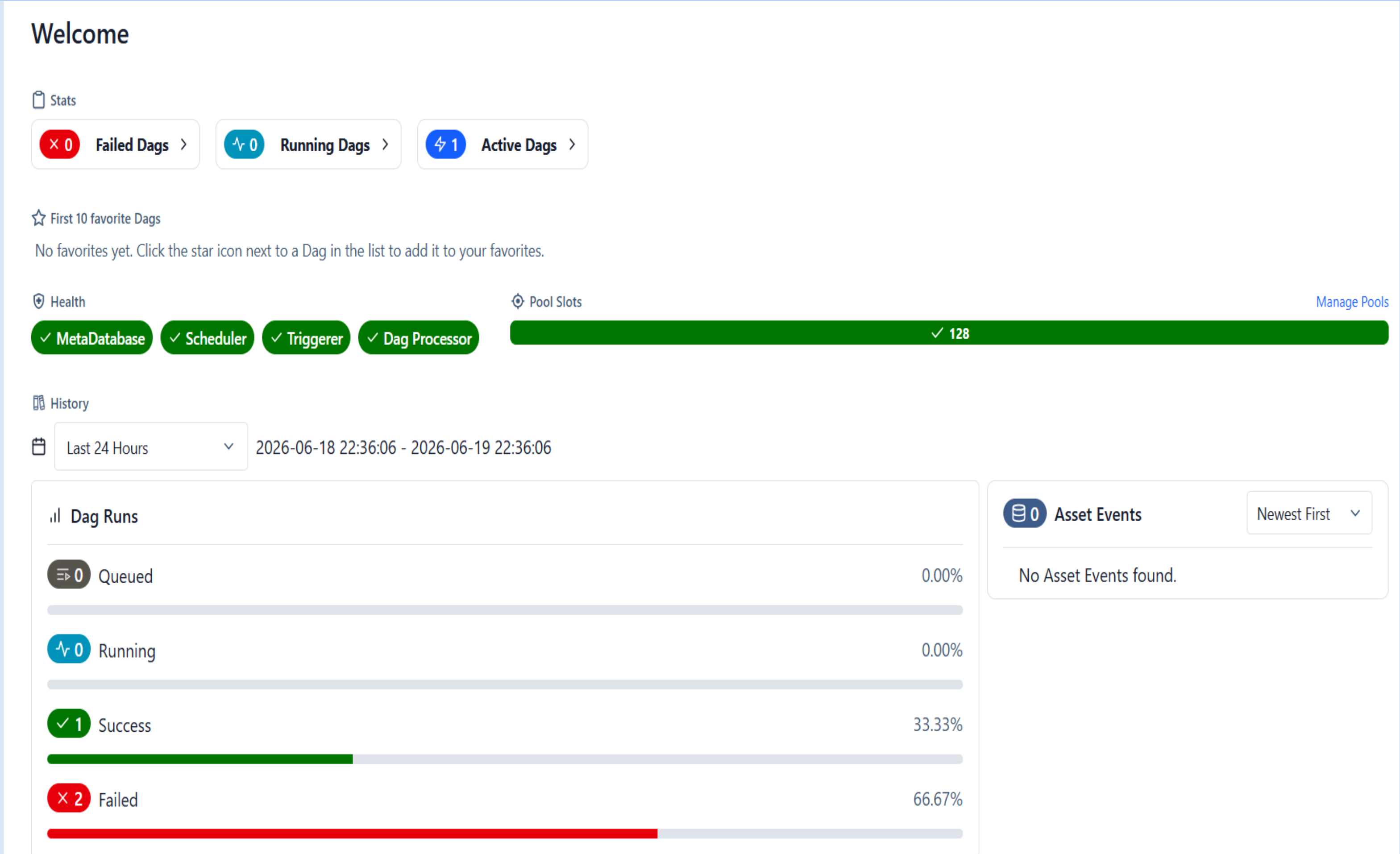


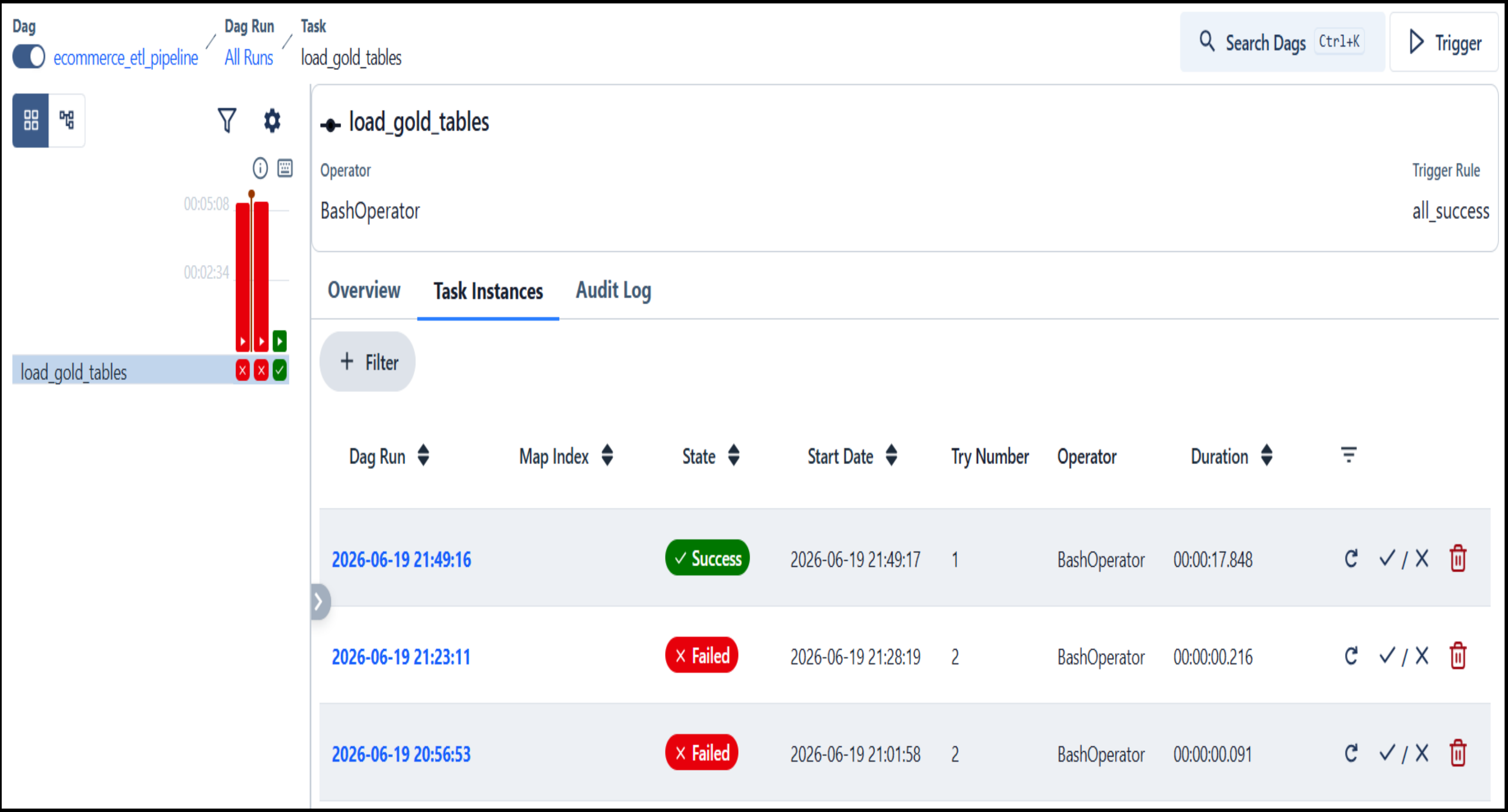
APACHE AIRFLOW

(Automating the ETL Pipeline Workflow)

Key Features

- Workflow Automation
- DAG Scheduling
- Task Monitoring
- Logging & Retry

Airflow automates the execution of the ETL pipeline, ensuring reliable scheduling, monitoring, and logging.



DAG EXECUTION

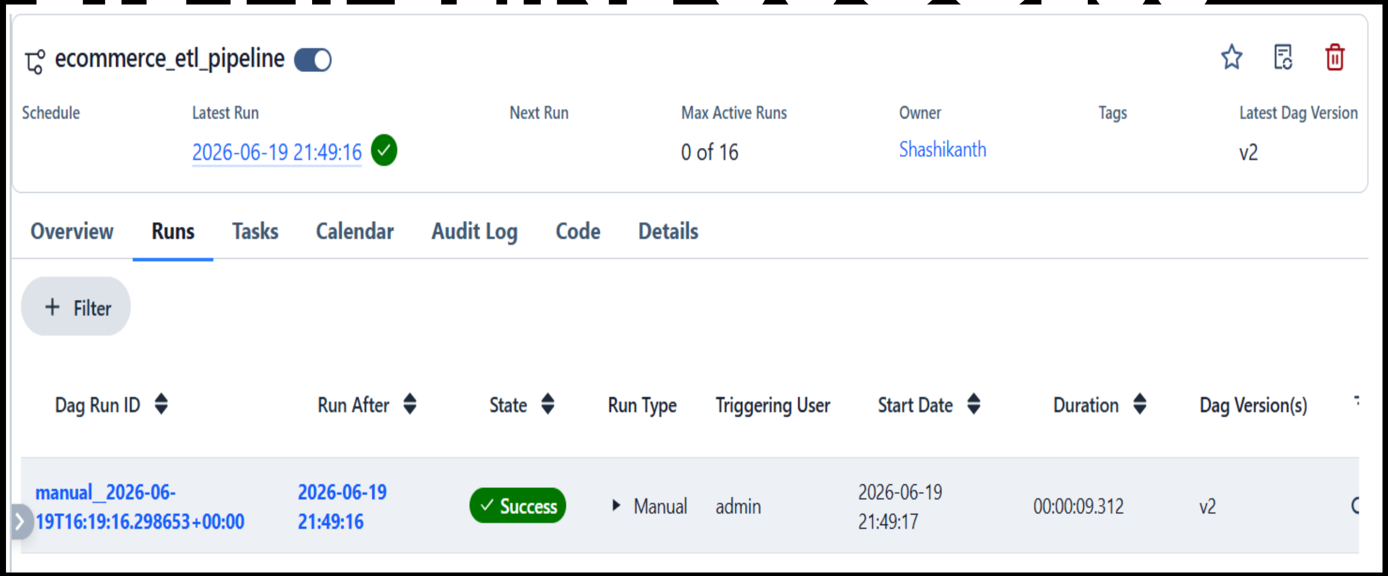
ETL Pipeline Execution

Workflow Steps

① DAG Triggered
② Read Gold Layer CSV Files
③ Execute `load_to_sqlite.py`
④ Load Data into SQLite
⑤ Task Completed Successfully

The Airflow DAG successfully executed the ETL workflow by loading the Gold Layer datasets into the SQLite database with execution monitoring and task status tracking.

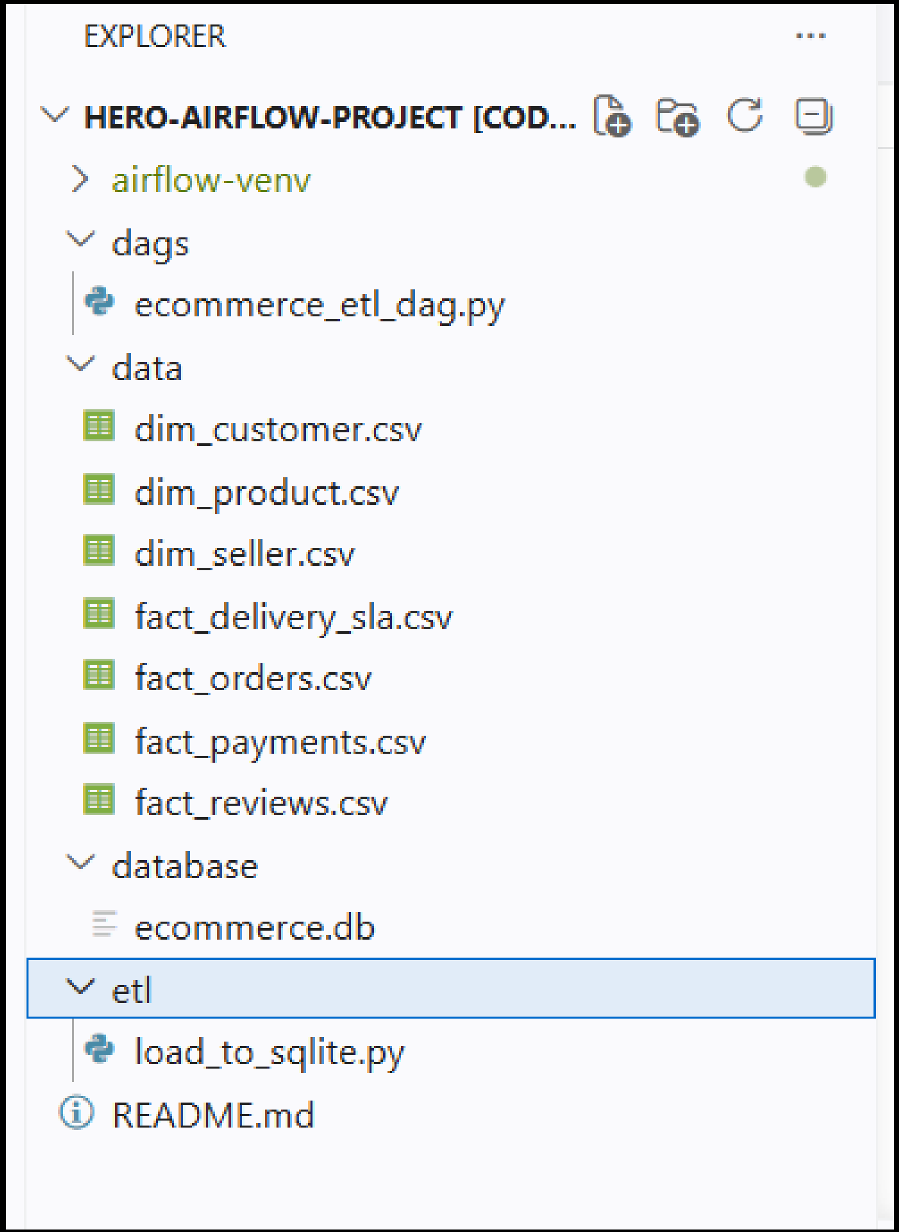
ETL PIPELINE SUCCESS



- ✅ ETL Pipeline Completed Successfully
- 📁 Gold Layer CSV Files
- |
- ▼
- Apache Airflow DAG
- |
- ▼
- 🗄 SQLite Database
- |
- ▼
- 📊 Power BI Dashboard

- DAG Executed Successfully
- Gold Layer Loaded
- SQLite Database Updated
- Dashboard Ready

The ETL pipeline completed successfully, transforming analytics-ready Gold Layer datasets into the SQLite database for business intelligence reporting through Power BI.



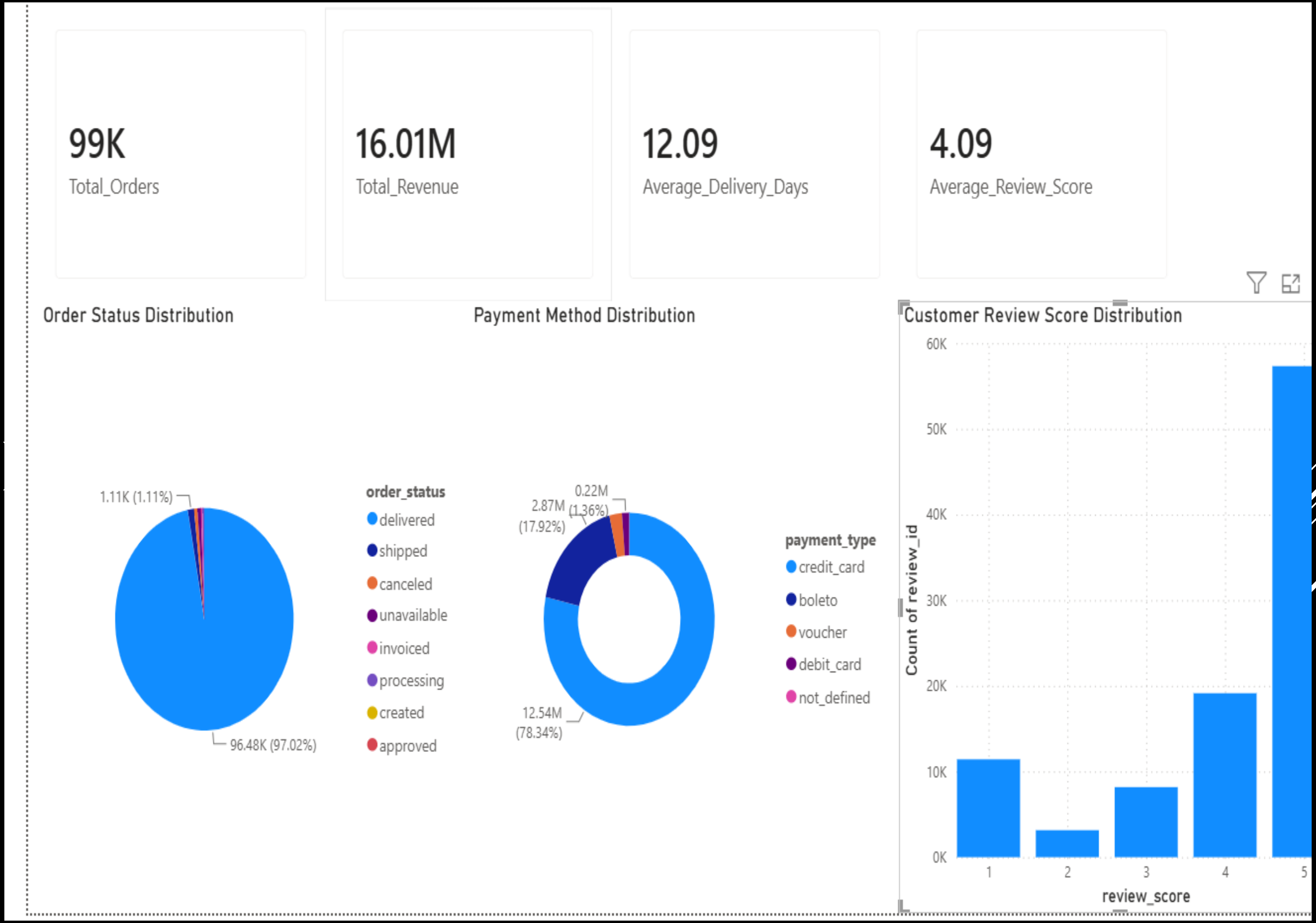
SQLITE DATABASE (Centralized Storage for Gold Layer Tables)

- 📋 **Database Features**
- • Lightweight Database
- • Stores Gold Layer Tables
- • Analytics Ready
- • Source for Power BI

💡 SQLite stores the transformed Gold Layer tables, providing a centralized data source for Power BI.

POWER BI DASHBOARD

(Interactive Business Intelligence & Performance Analytic



BUSINESS INSIGHTS

(Key Findings from the Power BI Dashboard)

📦 Order Performance

99K+ Orders were successfully processed through the ETL pipeline, providing a comprehensive view of order fulfilment performance.

💰 Revenue Analysis

The platform generated approximately **16.01 Million** in total revenue, indicating strong sales performance.

🚚 Delivery Performance

The average delivery time was **12.09 days**, reflecting an efficient order fulfilment process.

⭐ Customer Satisfaction

The average customer review score was **4.09/5**, showing a high level of customer satisfaction.

💳 Payment Insights

Credit Card was identified as the most preferred payment method among customers.

🎯 Business Value

The interactive dashboard enables stakeholders to monitor KPIs, evaluate operational performance, and make data-driven business decisions.

CHALLENGES & SOLUTIONS

(Challenges Encountered During ETL Pipeline Development)

🔹 **Challenge 1**
Multiple Raw Data Sources
Solution
Integrated multiple CSV datasets into a unified ETL pipeline using Python and Pandas.

🔹 **Challenge 2**
Data Quality Issues
Solution
Performed duplicate removal, timestamp conversion, missing value handling, and validation checks during the Silver Layer transformation.

🔹 **Challenge 3**
Workflow Automation
Solution
Implemented Apache Airflow to automate ETL execution, scheduling, monitoring, and logging.

🔹 **Challenge 4**
Centralized Data Storage
Solution
Stored analytics-ready Gold Layer tables in SQLite, creating a centralized repository for Power BI reporting.

💡 **Key Achievement**
Successfully developed an end-to-end automated Data Engineering pipeline that transforms raw e-commerce data into business-ready insights.

FUTURE SCOPE

(Potential Enhancements for the ETL Pipeline)

☁️ **Cloud Deployment**
Deploy the ETL pipeline on cloud platforms such as Microsoft Azure, AWS, or Google Cloud Platform for improved scalability.

⚡ **Big Data Processing**
Integrate Apache Spark to process large-scale datasets with better performance and distributed computing.

📡 **Real-Time Data Processing**
Implement real-time data ingestion and streaming using Apache Kafka or Apache Spark Streaming.

🤖 **Predictive Analytics**
Apply Machine Learning models to forecast sales, predict delivery delays, and analyze customer behavior.

🔄 **Incremental Data Loading**
Replace full data loads with incremental ETL processing to improve execution efficiency and reduce processing time.

🎯 **Long-Term Vision**
Develop a fully automated, cloud-based, real-time Data Engineering platform capable of supporting enterprise-scale analytics.

🎯 CONCLUSION

✅ KEY ACHIEVEMENTS

✓ End-to-End ETL Pipeline

✓ Medallion Architecture

✓ Apache Airflow Automation

✓ SQLite Data Storage

✓ Interactive Power BI Dashboard

💡 Final Takeaway

"Raw Data → Business Insights"

THANK YOU